Extras – maths, exam technique and mechanisms

X.1 Maths skills for the exam

Spec: Appendix 1: confident use of number, percentages and percentiles; ratios; surface areas and volumes; trigonometry; graphs and charts; coordinates and geometry; statistics and probability

Key facts

- **Area of a circle and circumference?** $A = \pi r^2$; $C = 2\pi r$ (or πd).
- **Volume of a cylinder?** $V = \pi r^2 h$.
- **Volume of a sphere?** $V = 4/3\ \pi r^3$ ($r = \sqrt[3]{(3V/4\pi)}$).
- **Volume of a frustum of a cone?** $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$.
- **How many mm² in 1 m²?** 1 000 000 mm² (1000 × 1000).
- **How many litres in 1 m³?** 1000 litres.
- **Mean from a frequency table?** Σ(value × frequency) ÷ Σfrequency.
- **When to round up in costing?** When buying whole packs, cans, sheets or containers – you can't buy part of a pack (e.g. 7.2 cans → 8).
- **Percentage of a total?** (part ÷ total) × 100 – make sure the total is the NEW total if values changed.
- **Hourly output formula?** O = P ÷ T, so P = O × T.

Skill	Past paper examples
Percentages, costs, waste allowances, profit	2021 Q4c bracket cost, 2024 Q2b glue waste, 2024 Q6e PPE %, 2025 Q2b rivets
Whole packs/cans/containers (round up)	2019 Q5e board packs, 2023 Q2b ink, 2024 Q5b paint
Area of compound shapes	2020 Q7 steel plate, 2024 Q5a floor, 2025 Q5a beam section
Surface area and volume	2019 Q4c varnish, 2020 Q2b frustum, 2022 Q2b milled block, 2022 Q5 sphere
Density and mass	2022 Q5b, 2025 Q5b
Pythagoras / trigonometry	2019 Q2b rafter
Statistics: mean, median, mode, probability	2020 Q4d, 2021 Q5e, 2023 Q5b
Graphs and pie charts	2020 Q6e extrapolation, 2023 Q5a pie chart
Rates and time	2021 Q2b ore reserves, 2023 Q6e fuel, 2025 Q6e hourly output
Nesting/maximum number of pieces	2022 Q6e cushion covers

Shape	Area	Volume
Rectangle / cuboid	l × w	l × w × h
Triangle / prism	½ × b × h	½ × b × h × length
Circle / cylinder	πr^2 (circumference $2\pi r$)	$\pi r^2 h$
Sphere	$4\pi r^2$	$4/3\ \pi r^3$
Cone	–	$\frac{1}{3}\pi r^2 h$
Frustum	–	$\frac{1}{3}\pi h(R^2 + Rr + r^2)$

Key terms

Percentage increase	New = original × (1 + %/100). E.g. 500 sticks + 30% = 500 × 1.3 = 650.
Pythagoras	In a right-angled triangle, hypotenuse² = a² + b².
SOH CAH TOA	sin θ = opp/hyp, cos θ = adj/hyp, tan θ = opp/adj.
Mean / median / mode	Mean = total ÷ number of values. Median = middle value when ordered. Mode = most frequent value.
Density	Density = mass ÷ volume, so mass = volume × density.
Unit conversion	1 m = 1000 mm; 1 m² = 1 000 000 mm²; 1 m³ = 1 000 000 000 mm³ = 1000 litres; 1 litre = 1000 cm³; 1 tonne = 1000 kg.
Probability of independent events	P(A and B) = P(A) × P(B).
Pie chart angle	Sector angle = (frequency ÷ total) × 360°.

X.2 Exam technique for Paper 1

Spec: Component 1: Principles of Design and Technology (9DT0/01) – 2 hours 30 minutes, 120 marks, 50% of the A level. All questions compulsory.; Command words: Give/State/Name, Describe, Describe using sketches, Calculate, Explain, Outline, Construct/Draw, Discuss/Consider, Justify, Analyse, Evaluate

Key facts

- **How is a 3-mark 'explain one' question marked?** 1 mark for identifying the point + 1 for a linked justification + 1 for a further linked consequence.
- **How is a 6-mark 'explain two' question marked?** Two separate points, each with two linked justifications (1+1+1 each). Don't repeat the same justification.
- **What do levels-based (6/9/12) answers need?** A wide range of relevant points, linked together and applied to the context in the question; for 'evaluate', a balanced judgement and clear conclusion.
- **Time per mark?** 150 minutes ÷ 120 marks ≈ 1.25 minutes per mark (a 12-mark question ≈ 15 minutes).
- **Common reasons marks are lost?** Repeating examples given in the stem, giving one point when two are needed, being too general, not linking to the product, writing about the wrong stage, not showing working, wrong rounding.
- **How to answer a design movement question?** Name key designers/dates, explain philosophy and style features, and link each feature to what you can see in the image.

Q	Typical content	Marks
1	Material/product context: give (2), explain two properties (4), explain one advantage (3)	9
2	Short recall (2) + calculation (5)	7
3	Material property (2) + describe a process with labelled sketches (4) + explain two benefits/drawbacks (6)	12
4	Industry/systems: give/outline (2–6) + a 9-mark discuss	20
5	Calculations	8
6	Mixed: H&S, production, textiles, QA + calculation	19
7	Drawing: isometric / orthographic / perspective / net	6
8	Design movement (discuss, 9)	9
9	Discuss (9) – technology, systems, ergonomics	9
10	Smart/modern material: explain three uses / two benefits and one drawback (9)	9
11	Evaluate a product (12)	12

Command	What to do	Mark pattern
Give / State / Name	Brief recall	1 per point
Describe (using sketches)	Stages in correct order; labelled sketches	1 per stage; capped without sketches/labels or with wrong order
Calculate	Formula, substitution, working, units, rounding	Stage (method) marks + ECF
Explain	Point → because/so → which means…	2 = 1+1, 3 = 1+1+1
Outline	Clear sequence of key points/stages	1 per point
Discuss	Range of factors, linked, applied to context	Levels (6: 3 levels; 9: 3 levels)
Evaluate	Analyse, weigh strengths/weaknesses, supported **conclusion**	Levels (9 or 12 – 12 has 4 levels)

Key terms

Give / State / Name	Recall – short answers, one mark per correct point. No explanation needed.
Describe	Give an account of a process in the correct order, or a series of features. Number of points = number of marks.
Describe using sketches	Labelled/annotated sketches showing stages in order. Without sketches or labels marks are capped (often max 2–3).
Calculate	Show all working; method marks for each stage, error carried forward.
Explain	Point + linked reason/justification. 2 marks = point + one link; 3 marks = point + two links.
Outline	Clear, coherent sequence of essential features or stages (6 marks = 6 points). No conclusion needed.
Discuss	Explore the issue from different angles with reasoning and evidence, using subject terminology. No conclusion required – levels marked.

Evaluate	Critically review strengths and weaknesses using evidence and reach a supported conclusion – levels marked.

X.3 Mechanisms – levers, linkages, cams and gears

Spec: Extension topic: not listed in the Edexcel 9DT0 Paper 1 content, but useful background for analysing products (Q11) and for NEA prototypes. Builds on GCSE systems and mechanical devices.

Key facts

• **Four types of motion?** Rotary, linear, reciprocating, oscillating.
• **Give an example of each lever class.** Class 1: see-saw, scissors, crowbar. Class 2: wheelbarrow, nutcracker, bottle opener. Class 3: tweezers, fishing rod, human forearm.
• **MA formula?** MA = load ÷ effort.
• **Driver 20 teeth, driven 60 teeth, input 300 rpm. Output speed?** Ratio 60:20 = 3:1, output = 300 ÷ 3 = 100 rpm (opposite direction).
• **Which cam gives a sudden drop?** Snail (drop) cam – can only rotate one way.
• **Which cam gives uniform rise and fall?** Heart-shaped cam.
• **What does a reverse motion linkage do?** Output moves in the opposite direction to the input (fixed pivot in the middle).
• **What does a bell crank do?** Changes the direction of motion through 90° (e.g. bicycle brakes).
• **Crank and slider converts?** Rotary to reciprocating motion (or vice versa) – engine piston, pumps.
• **Worm and wheel features?** Very high ratio (e.g. 40:1 with a 40-tooth wheel), changes axis by 90°, cannot be back-driven.
• **Rack and pinion converts?** Rotary to linear motion – car steering, pillar drill feed.

Class	Order	Examples	MA
1	Effort – **Fulcrum** – Load	See-saw, crowbar, scissors, pliers	Can be > or < 1
2	Effort – **Load** – Fulcrum	Wheelbarrow, nutcracker, bottle opener	Always > 1
3	**Effort** in the middle	Tweezers, tongs, fishing rod, forearm	Always < 1 (gains distance/speed)

Key terms

Mechanism	A device that changes an input force and motion into a desired output force and motion.
Types of motion	Rotary (round), linear (straight line), reciprocating (backwards and forwards in a line), oscillating (swinging in an arc).
Lever	A rigid bar that pivots about a fulcrum. Class 1: fulcrum in the middle; class 2: load in the middle; class 3: effort in the middle.
Mechanical advantage (MA)	Load ÷ effort. MA > 1 means the mechanism magnifies force.
Velocity ratio (VR)	Distance moved by effort ÷ distance moved by load.
Linkage	Levers joined together to change the direction or size of motion – reverse motion, parallel motion, bell crank, crank and slider, treadle.
Cam and follower	A rotating cam converts rotary motion into reciprocating or oscillating motion of a follower. Profiles: circular (eccentric), pear, heart, snail (drop).
Gear ratio	Number of teeth on driven gear ÷ number of teeth on driver gear. Output speed = input speed ÷ gear ratio.
Idler gear	A gear between driver and driven that makes them rotate in the same direction without changing the ratio.
Compound gear train	Two gears fixed on the same shaft, giving a large overall ratio = product of the individual ratios.

Exam tip: Note: Mechanisms are **not** named in the Edexcel 9DT0 Paper 1 subject content, so they are unlikely to be examined directly. They are included here because they help you analyse products in 12-mark evaluations and are often needed in NEA prototypes.